

Nonlinear Systems and Coupled Models

Differentiable Physical Modelling and Machine Learning for Sound Synthesis

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2026-05-21

Week 3 – Nonlinear Systems and Coupled Models

1

Overview

- Recap: diagonal modal dynamics and complex poles
- Activities: modal recursion, tension modulation, pole recovery
- Nonlinearities as mode couplings
- Geometric nonlinearity: state-dependent tension
- Brief outlook: contact, excitation, and coupled bodies
- Differentiability: estimate parameters and forces

Recap: Linear Modal Systems

Week 2

We wrote the displacement field as a sum of mode shapes:

$$w(x, t) = \sum_{\mu=1}^N \phi_{\mu}(x) q_{\mu}(t)$$

- $\phi_{\mu}(x)$ is the shape of mode μ
- $q_{\mu}(t)$ is its time-varying amplitude
- In a linear system, each mode behaves like its own oscillator

$$\ddot{q}_{\mu} + 2\gamma_{\mu} \dot{q}_{\mu} + \omega_{\mu}^2 q_{\mu} = b_{\mu} f(t)$$

From Oscillator to Poles

First ignore the input:

$$\ddot{q}_\mu + 2\gamma_\mu \dot{q}_\mu + \omega_\mu^2 q_\mu = 0$$

Assume an exponential solution:

$$q_\mu(t) = e^{s_\mu t}$$

Substitute into the oscillator equation:

$$s_\mu^2 + 2\gamma_\mu s_\mu + \omega_\mu^2 = 0$$

So each mode has two complex poles:

$$s_{\mu,\pm} = -\gamma_\mu \pm i\sqrt{\omega_\mu^2 - \gamma_\mu^2}$$

From Oscillator to Poles

The exponential form makes differentiation simple:

$$q_\mu(t) = e^{s_\mu t}$$

$$\dot{q}_\mu(t) = s_\mu e^{s_\mu t}, \quad \ddot{q}_\mu(t) = s_\mu^2 e^{s_\mu t}$$

Substitute into the oscillator:

$$(s_\mu^2 + 2\gamma_\mu s_\mu + \omega_\mu^2) e^{s_\mu t} = 0$$

Since $e^{s_\mu t} \neq 0$, the pole equation is:

$$s_\mu^2 + 2\gamma_\mu s_\mu + \omega_\mu^2 = 0$$

The Laplace transform formalises the same idea: differentiation in time becomes multiplication by s .

Each Mode as a Complex State

Because the oscillator has real coefficients, the two poles are complex conjugates:

$$s_\mu = -\gamma_\mu + i\omega_{d,\mu}, \quad s_\mu^* = -\gamma_\mu - i\omega_{d,\mu}$$

Store one complex modal amplitude $\tilde{q}_\mu$:

$$\dot{\tilde{q}}_\mu = s_\mu \tilde{q}_\mu + \tilde{b}_\mu f(t)$$

The conjugate trajectory follows automatically:

$$\dot{\tilde{q}}_\mu^* = s_\mu^* \tilde{q}_\mu^* + \tilde{b}_\mu^* f(t)$$

The real modal displacement is recovered by taking the real part:

$$q_\mu(t) = \text{Re}\{\tilde{q}_\mu(t)\}$$

For light damping:

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$$\omega_{d,\mu} = \sqrt{\omega_\mu^2 - \gamma_\mu^2} \approx \omega_\mu$$

Linear Modes Do Not Talk

Stack the stored complex amplitudes into one vector:

$$\dot{\tilde{\mathbf{q}}} = \mathbf{A}\tilde{\mathbf{q}} + \tilde{\mathbf{b}}f(t)$$

For a linear modal system, the dynamics matrix is diagonal:

$$\mathbf{A} = \begin{bmatrix} s_1 & 0 & 0 & \dots \\ 0 & s_2 & 0 & \dots \\ 0 & 0 & s_3 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

The zeros off the diagonal mean:

mode μ does not directly affect mode ν

Week 3 starts when these zeros stop being zeros.

Mini Task: Make the Diagonal System Sound

Activity: Complex Modal Recursion

Notebook: [week03/notebooks/complex_modal_recursion.py](#)

You are given:

$$\boldsymbol{\omega} = [\omega_1 \quad \omega_2 \quad \dots \quad \omega_N]^\top, \quad \boldsymbol{\gamma} = [\gamma_1 \quad \gamma_2 \quad \dots \quad \gamma_N]^\top$$

and an initial complex modal state:

$$\tilde{\mathbf{q}}^0 \in \mathbb{C}^N$$

In the notebook, this complex state is named $\mathbf{z}$.

Build the continuous-time diagonal system:

$$\mathbf{A} = \text{diag}(\mathbf{s}), \quad s_\mu = -\gamma_\mu + i\omega_\mu$$

Then discretise it:

$$\tilde{\mathbf{q}}^{n+1} = \text{diag}(\mathbf{a}) \tilde{\mathbf{q}}^n, \quad a_\mu = e^{s_\mu \Delta t}$$

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Finally produce the audio output:

$$y^n = \operatorname{Re} \left\{ \sum_{\mu=1}^N c_{\mu} \tilde{q}_{\mu}^n \right\}$$

Activity: Three Ways to Advance the State

The exact solution is a matrix exponential:

$$\tilde{\mathbf{q}}^n = e^{\mathbf{A}n\Delta t} \tilde{\mathbf{q}}^0$$

Since $\mathbf{A}$ is diagonal:

$$\tilde{\mathbf{q}}^n = \text{diag}(\mathbf{a})^n \tilde{\mathbf{q}}^0, \quad a_\mu = e^{s_\mu \Delta t}$$

So the state advance is just a cumulative product. All methods should produce the same sound:

1. Cumulative product

$$\tilde{q}_\mu^n = \tilde{q}_\mu^0 \prod_{k=0}^{n-1} a_\mu$$

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2. Vandermonde / direct evaluation

$$\tilde{q}_{\mu}^n = \tilde{q}_{\mu}^0 a_{\mu}^n$$

3. Sequential recursion

$$\tilde{\mathbf{q}}^{n+1} = \mathbf{a} \odot \tilde{\mathbf{q}}^n$$

4. Parallel scan

Use `jax.lax.associative_scan` over the multiplicative updates.

Skeleton:

```

1 sr = 44100
2 dt = 1 / sr
3 n_steps = 2 * sr
4
5 s = -gamma + 1j * omega
6 a = jnp.exp(s * dt)
7
8 # TODO: compute z over time, shape (n_steps, n_modes)
9 # TODO: y = real(z @ c)
10 # TODO: normalise and listen

```

Checkpoint question:

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Nonlinearities and Mode Coupling

What Breaks the Diagonal System?

In the linear case:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q}$$

with diagonal $\mathbf{A}$.

With nonlinear physics:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{nl}(\mathbf{q})$$

The diagonal part still describes the independent modal resonances.

The new term $\mathbf{nl}(\mathbf{q})$ collects the nonlinear state-dependent forces.

The matrix can stay diagonal, but the dynamics are no longer decoupled.

Nonlinearities as Mode Couplings

In modal coordinates, the nonlinear term becomes a combination of modal amplitudes.

Today we mostly care about **cubic** couplings.

For tension modulation, the cubic has a special structure:

$$\mathbf{nl}_\mu(\mathbf{q}) \sim \sum_\nu q_\mu q_\nu q_\nu$$

One mode q_μ is multiplied by a global quadratic stretch:

$$\sum_\nu q_\nu^2$$

For a more general geometric nonlinearity, such as a plate, all three modal indices can be different:

$$\mathbf{nl}_\mu(\mathbf{q}) = \sum_{i,j,k} C_{ijk} q_i q_j q_k$$

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In code:

```
1 stretch = jnp.einsum("n->", q**2)
2 nl_tension = q * stretch
3
4 nl_plate = jnp.einsum("mijk,i,j,k->m", C, q, q, q)
```

Next: derive the structured tension case from the wave equation.

Physical Sources of Nonlinearity

- Main focus today: geometric nonlinearity
- A vibrating string stretches as it moves
- Stretching increases the effective tension
- Increased tension changes the modal frequencies
- Since the tension depends on the current state, the modes become coupled

Other nonlinearities exist too: contact, collisions, bow friction, reeds, hammers.

But today we mostly follow one thread:

$$\dot{\mathbf{q}} = \mathbf{A}\mathbf{q} + \mathbf{r}_{\text{geom}}(\mathbf{q}, \theta)$$

Week 3 Roadmap

1. Mode coupling from nonlinear terms
2. Kirchhoff–Carrier string and tension modulation
3. Energy transfer between modes
4. Brief outlook: contact, excitation and coupled bodies
5. Why differentiability helps estimate nonlinear parameters

Geometric Nonlinearity

The String Gets Longer

A displaced string is slightly longer than a straight string.

For small slopes, the extra length is approximately:

$$\Delta L(t) \approx \frac{1}{2} \int_0^L (\partial_x w(x, t))^2 dx$$

- the extra length depends on the current shape
- larger slopes produce more stretching
- stretching changes the tension

State-Dependent Tension

The increased length raises the string tension:

$$T(w) = T_0 + \frac{EA}{2L} \int_0^L (\partial_x w(x, t))^2 dx$$

- T_0 : resting tension
- E : Young's modulus
- A : cross-sectional area
- $T(w)$ is a global scalar, uniform along the string
- the string becomes stiffer when it vibrates more strongly

This is the source of the nonlinearity:

$$T = T(w)$$

Here w means the whole current string shape. Time dependence enters through $w(x, t)$.

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Wave Equation with f_{nl}

Keep the linear string on the left and isolate the nonlinear force:

$$\rho A \partial_t^2 w = T_0 \partial_x^2 w + f_{\text{ext}} - f_{\text{nl}}(w)$$

In modal form:

$$\ddot{q}_\mu + 2\gamma_\mu \dot{q}_\mu + \omega_\mu^2 q_\mu = \frac{1}{\rho} (\bar{f}_{\text{ext},\mu} - \bar{f}_{\text{nl},\mu}(\mathbf{q}))$$

Now focus only on:

$$\bar{f}_{\text{nl},\mu}(\mathbf{q})$$

Kirchhoff-Carrier Nonlinear Force

With the sign convention from the previous slide, $f_{\text{nl}}(w)$ is the nonlinear restoring force:

$$f_{\text{nl}}(w) = -T_{\text{nl}}(w) \partial_x^2 w$$

with

$$T_{\text{nl}}(w) = \frac{EA}{2L} \int_0^L (\partial_x w)^2 dx$$

So:

$$f_{\text{nl}}(w) = - \left[\frac{EA}{2L} \int_0^L (\partial_x w)^2 dx' \right] \partial_x^2 w$$

Projected Nonlinear Force

Start from:

$$f_{\text{nl}}(w) = - \left[\frac{EA}{2L} \int_0^L (\partial_x w)^2 dx' \right] \partial_x^2 w$$

Using the modal expansion and orthonormal simply supported modes:

$$\bar{f}_{\text{nl},\mu}(\mathbf{q}) = \left(\frac{EA}{2L} \sum_{\nu} \lambda_{\nu}^2 q_{\nu}^2 \right) \lambda_{\mu}^2 q_{\mu}$$

What happened during projection:

$$\int_0^L \phi_i' \phi_j' dx \rightarrow \begin{cases} \lambda_i^2, & i = j \\ 0, & i \neq j \end{cases}$$

$$\int_0^L \phi_i'' \phi_{\mu} dx \rightarrow \begin{cases} -\lambda_{\mu}^2, & i = \mu \\ 0, & i \neq \mu \end{cases}$$

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So the explicit minus in $f_{\text{nl}}(w)$ cancels the minus from the curvature projection.

Modal Kirchhoff-Carrier Equation

Use the same force convention as before:

$$\ddot{q}_\mu + 2\gamma_\mu \dot{q}_\mu + \omega_\mu^2 q_\mu = \frac{1}{\rho} (\bar{f}_{\text{ext},\mu} - \bar{f}_{\text{nl},\mu}(\mathbf{q}))$$

$$\bar{f}_{\text{nl},\mu}(\mathbf{q}) = \frac{EA}{2L} \lambda_\mu^2 q_\mu \sum_\nu \lambda_\nu^2 q_\nu^2$$

With no external force, the modal ODE is:

$$\ddot{q}_\mu + 2\gamma_\mu \dot{q}_\mu + \omega_\mu^2 q_\mu + \frac{1}{\rho} \bar{f}_{\text{nl},\mu}(\mathbf{q}) = 0$$

Structured Cubic Coupling

Each modal force contains terms like:

$$q_\mu q_1^2, \quad q_\mu q_2^2, \quad q_\mu q_3^2, \quad \dots$$

Ignoring physical constants for a moment:

```
1 stretch = jnp.einsum("n->", q**2)
2 n1 = q * stretch
```

This is cubic, but it is not a dense cubic tensor:

one global stretch scalar $\times$ one modal displacement

Physical Consequence

When the vibration amplitude is large:

- the string stretches more
- the effective tension increases
- the modal frequencies move upward
- as the sound decays, the tension relaxes
- the pitch falls back toward the linear value

This is often called **tension modulation**.

It is one of the simplest ways a linear modal string becomes nonlinear.

Mini Task: Hear the Nonlinearity

Activity: Tension Modulation

Notebook: [week03/notebooks/nonlinear_tension_modulation.py](#)

Start with the linear modal equation:

$$\ddot{q}_\mu = -2\gamma_\mu \dot{q}_\mu - \omega_\mu^2 q_\mu$$

Then add one structured cubic restoring force:

$$\ddot{q}_\mu = -2\gamma_\mu \dot{q}_\mu - \omega_\mu^2 q_\mu - \beta \lambda_\mu^2 q_\mu \left(\sum_\nu \lambda_\nu^2 q_\nu^2 \right)$$

In the notebook, `lambda_mu` stores the squared wavenumber from `jaxdiffmodal`.

The TODO is just:

```
1 stretch = jnp.einsum("n,n->", lambda_mu, q**2)
2 force = beta * lambda_mu * q * stretch
```

Ask students to compare:

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- $\beta = 0$ versus $\beta > 0$
- small versus large pluck amplitude
- waveform shape during the attack and decay
- whether the perceived pitch stays fixed

What Students Should Notice

- The nonlinear model is amplitude-dependent.
- The effect is strongest near the attack.
- As the vibration decays, the effective tension relaxes.
- A linear system can get louder or quieter, but its modal frequencies do not depend on the current state.
- The coupling is compact: one scalar **stretch** affects every modal equation.

Other Nonlinear Forces

Contact and Collisions

Some nonlinearities switch on only when two objects touch.

A simple contact law:

$$F_c(y) = k [y - y_c]_+^\alpha$$

where

$$[x]_+ = \max(x, 0)$$

- no force before contact
- rapidly increasing force after contact
- examples: hammer-string, mallet-membrane, rattling objects

This is local in physical space, but becomes mode coupling after projection.

Nonlinear Excitation

Some instruments are nonlinear mainly at the excitation point.

Examples:

- **bowed string:** friction depends on relative velocity
- **reed instruments:** airflow depends on pressure and reed opening
- **hammered string:** felt compression creates a nonlinear contact force

Common pattern:

$$F_{\text{exc}} = F(y, \dot{y}, u; \theta)$$

The resonator can still be modal, but the force law is state-dependent.

Coupled Systems

String + Body

A guitar is not just a string.

The string injects force into the bridge, and the body moves in response:

string $\longleftrightarrow$ bridge $\longleftrightarrow$ body

Each subsystem can still be modal:

$$w_s(x, t) = \sum_{\mu} \phi_{\mu}^s(x) q_{\mu}^s(t), \quad w_b(\mathbf{x}, t) = \sum_{\nu} \phi_{\nu}^b(\mathbf{x}) q_{\nu}^b(t)$$

How do they interact? Let's look at three levels of complexity.

Model 1: The One-Way Street (Strict Decoupling)

What if the string assumes the bridge is a brick wall?

The string vibrates against a rigid boundary. It doesn't "feel" the body yielding:

$\ddot{\mathbf{q}}_s$ = string dynamics (isolated)

However, the bridge still passes the resulting force to the body. The body acts simply as a filter:

$\ddot{\mathbf{q}}_b$ = body dynamics + $\mathbf{b}_b F_{string}$

Model 2: The Two-Way Street (Boundary Impedance)

Real bridges yield. The strings and body form a two-way feedback loop.

Let the bridge displacement be read from each modal system:

$$y_s = \mathbf{c}_s^\top \mathbf{q}_s, \quad y_b = \mathbf{c}_b^\top \mathbf{q}_b$$

The bridge acts as a mechanical impedance (a coupling force):

$$F_c = k_c(y_s - y_b) + r_c(\dot{y}_s - \dot{y}_b)$$

The same interaction force acts on both, with opposite signs:

$$\ddot{\mathbf{q}}_s = \text{string dynamics} - \mathbf{b}_s F_c$$

$$\ddot{\mathbf{q}}_b = \text{body dynamics} + \mathbf{b}_b F_c$$

Model 3: Sympathetic Resonance (Impedance Network)

A guitar has more than one string sharing the exact same bridge.

Because the bridge (y_b) is shared, energy flows back up into the other strings:

String A $\longrightarrow$ Bridge $\longrightarrow$ String B

The single coupling force F_c expands into an **impedance network**.

- Pluck one string, the others ring.
- Modal banks remain individually simple to compute.
- The interaction network creates the complex reality.

Differentiability

Start with the Linear Case

Before learning nonlinear forces, start with the linear modal model:

$$\tilde{q}_\mu(t) = \tilde{q}_\mu(0) e^{(-\gamma_\mu + i\omega_\mu)t}$$

The output is:

$$\hat{y}(t) = \text{Re} \left\{ \sum_{\mu} c_{\mu} \tilde{q}_{\mu}(t) \right\}$$

Fit the parameters by minimising:

$$\mathcal{L} = \frac{1}{T} \sum_n (\hat{y}_n - y_n)^2$$

Differentiable Does Not Mean Easy

Can we differentiate through the model?

Yes, in the parameter range we use, the modal recursion, synthesis equation, and loss have well-defined gradients.

Are the gradients informative?

Sometimes.

- Good initialisation can point the optimiser towards the right basin.
- Poor initialisation can land in a local minimum.
- Frequency and phase errors can make the loss surface highly oscillatory.

What Are We Recovering?

The fitted poles are:

$$s_\mu = -\gamma_\mu + i\omega_\mu$$

These are not just abstract numbers.

For a stiff string:

$$\omega_\mu^2 = \frac{T_0}{\rho A} \lambda_\mu^2 + \frac{EI}{\rho A} \lambda_\mu^4$$

So the frequency curve contains information about:

- tension T_0
- density ρ
- cross-sectional area A
- stiffness EI

Damping Parameters

The decay rates also have physical structure:

$$2\gamma_\mu \approx d_1 + d_3 \lambda_\mu^2$$

So fitting γ_μ across modes can recover:

- frequency-independent damping d_1
- frequency-dependent damping d_3

This is the first differentiable inverse problem:

audio or modal trajectory $\rightarrow \omega_\mu, \gamma_\mu \rightarrow$ physical parameters

Activity: Recover the Poles

Notebook: [week03/notebooks/complex_modal_recursion.py](#)

Part 2 fits the poles of a target linear modal simulation:

$$s_\mu = -\gamma_\mu + i\omega_\mu, \quad \overline{\mathbf{A}} = \text{diag}(e^{s_\mu \Delta t})$$

Students choose the initial pole guesses and complete the stability constraint:

$$\text{Re}(s_\mu) \leq 0, \quad |\text{diag}(\overline{\mathbf{A}})_\mu| \leq 1$$

Then Optax updates the unconstrained parameters by gradient descent.

Then Add Nonlinear Forces

Once the linear part is identified:

$$\ddot{\mathbf{q}} = -2\Gamma\dot{\mathbf{q}} - \Omega^2\mathbf{q} - \mathbf{nl}(\mathbf{q})$$

For tension modulation, we know:

$$\mathbf{nl}_\mu(\mathbf{q}) = \frac{EA}{2\rho L} \lambda_\mu^2 q_\mu \sum_\nu \lambda_\nu^2 q_\nu^2$$

For more complex systems, we may learn:

$$\mathbf{nl}_\theta(\mathbf{q})$$

Differentiability lets us optimise physical parameters, initial conditions, or neural residuals through the simulator.

Week 3 Takeaway

Linear modal synthesis gives:

independent modes

Geometric nonlinearity gives:

state-dependent coupling

Differentiability lets us turn the simulator around:

sound $\rightarrow$ parameters and forces